

SYSTEMS AND METHODS FOR EXPLICIT
COMPOSITIONAL REASONING IN A
CONTINUOUS-PRESENT COGNITIVE
ARCHITECTURE

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SYSTEMS AND METHODS FOR EXPLICIT COMPOSITIONAL REASONING IN A CONTINUOUS-PRESENT COGNITIVE ARCHITECTURE

CROSS-REFERENCES TO RELATED APPLICATIONS

This application claims priority from Australian Provisional Patent Application No. 2025905860 and other related provisional filings made by the inventor in November 2025, which describe aspects of the Aurora reasoning architecture and the Persistent Existence Frame (PEF). These filings are incorporated herein by reference for all purposes permitted under applicable law.

The present application does **not** redefine or restate the PEF as disclosed in earlier filings.

Instead, it provides:

- the explicit machinery of compositional reasoning,
- the structural mechanisms required for interpretation, collapse, and ambiguity management,
- the complete definition of compositional primitives,
- the Aurora Block execution engine,
- and the integrated interaction between Aurora and the previously disclosed PEF.

FIELD OF THE INVENTION

The invention relates to artificial intelligence, machine reasoning, cognitive architectures, and structured semantic processing. More particularly, it concerns a non-temporal reasoning substrate using structured conceptual entities—Roles, Domains, and Spans—transformed through explicit compositional primitives, evaluated through interpretability metrics, and processed by a deterministic execution module.

The invention enables reasoning without reliance on token prediction, temporal sequence modeling, vector-space drift, or latent statistical inference.

BACKGROUND OF THE INVENTION

Machine reasoning systems today overwhelmingly rely on statistical inference. Large transformer-based models excel in fluency but lack:

- persistent conceptual identity,
- structured state representations,
- principled mechanisms for ambiguity maintenance,
- explicit operators for compositional reasoning,
- deterministic collapse logic,
- interpretable internal transitions,
- and non-temporal reasoning substrates.

These deficiencies lead to:

- premature interpretive collapse,
- invented grammatical or causal heuristics,
- hallucinated structures inconsistent with prior context,
- inability to maintain multiple competing interpretations,
- loss of identity across linguistic spans,
- incoherent state transitions.

Existing architectures attempt to mitigate these weaknesses through post-hoc guardrails, fine-tuning, or constrained decoding.

However, these approaches do not provide:

- persistent conceptual anchors,
- structural meaning-fields aligned with verbs and modality,
- bounded reasoning episodes independent of time,
- operator-level reasoning primitives,
- nor cross-compatible collapse logic.

Consequently, they cannot support stable interpretability, multi-interpretation maintenance, nor deterministic compositional reasoning.

There is therefore a need for:

- a reasoning substrate independent of temporal representation,

- a structured conceptual geometry,
- explicit compositional operators,
- deterministic primitive activation and collapse,
- interpretability-based state management,
- and a processing engine capable of updating meaning within a continuous present.

This invention provides such a system.

SUMMARY OF THE INVENTION

The invention provides a **non-temporal cognitive reasoning architecture** comprising:

- **Roles** (persistent conceptual entities),
- **Domains** (semantic meaning-fields activated by verbs, modality, or continuity cues),
- **Spans** (bounded reasoning episodes forming conceptual—not temporal—sequence),
- a family of **compositional primitives** (WE, THEN, WHILE, UNTIL, BECAUSE),
- an **Interpretability Engine** ($I = S + E - D$),
- **Domain Activation Logic**,
- a **Span Boundary Algorithm**,
- a **Parallel Interpretation Engine**,
- the **Aurora Block** (deterministic reasoning cycle),
- and integration with the **Persistent Existence Frame (PEF)** in applicable embodiments.

This architecture provides explicit, interpretable, deterministic reasoning without reliance on token-predictive models.

Conceptual State Representation

Roles serve as persistent anchors; Domains define allowable meaning transformations; Spans segment conceptual episodes.

Compositional Primitives

Each primitive is defined by:

- semantic function,
- invariants,
- preconditions,
- activation mechanics,
- transformation effects,
- collapse conditions.

These primitives enable:

- collective formation (WE),
- conceptual succession (THEN),
- conditional continuity (WHILE),
- goal-bound persistence (UNTIL),
- causal attribution (BECAUSE).

Interpretability Engine

Interpretability determines:

- Role compatibility,
- Domain stability,
- primitive lifecycle,
- span continuation or collapse,
- and ambiguity resolution.

Parallel Interpretation Engine

Multiple structurally valid interpretations persist simultaneously until collapse is forced by structure, not heuristics.

Aurora Block

A deterministic reasoning cycle:

1. Interpret input
2. Evaluate Span boundaries

3. Activate or update Domains
4. Update interpretability
5. Evaluate primitive eligibility
6. Apply primitives
7. Evaluate collapse
8. Update parallel interpretations
9. Emit updated state

PEF Integration

The architecture operates inside the PEF's continuous present, using Active Presence, Echo State, and Potential State for:

- reconstruction,
- projection,
- ambiguity maintenance,
- and structural persistence.

Together these elements form a complete, explicit reasoning engine.

ABSTRACT

This invention discloses a reasoning architecture employing structured conceptual entities—Roles, Domains, Spans—transformed through explicit compositional primitives executed within a continuous-present substrate. A family of primitives (WE, THEN, WHILE, UNTIL, BECAUSE) operate deterministically based on defined preconditions, invariants, activation rules, transformation effects, and collapse conditions.

An Interpretability Engine computes structural alignment via similarity, evidence, and decay. Domain Activation Logic determines semantic meaning-fields. A Span Boundary Algorithm regulates conceptual segmentation. A Parallel Interpretation Engine maintains multiple interpretations until structural conditions force collapse.

A deterministic execution module, the Aurora Block, orchestrates input interpretation, primitive activation, collapse evaluation, and state synthesis. In certain embodiments the architecture integrates with the previously disclosed Persistent Existence Frame (PEF) but does not redefine it.

The result is an explicit, interpretable, operator-driven reasoning system independent of token prediction and temporal sequence modeling.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1

High-Level System Architecture.

Shows the conceptual components of the reasoning system, including:

- Roles (101),
- Domains (102),
- Spans (103),
- Compositional primitives (120),
- Interpretability Engine (140),
- Domain Activation Logic (150),
- Span Boundary Algorithm (160),
- Parallel Interpretation Engine (170),
- Aurora Block execution module (180),
- and integration with the Persistent Existence Frame (PEF) substrate (190).

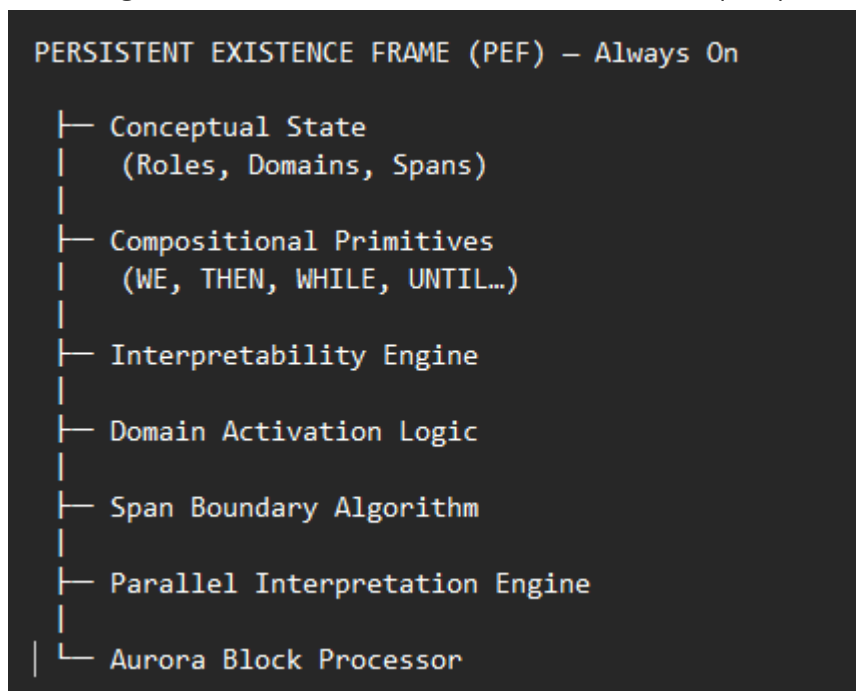


FIG. 2

Roles as Conceptual Anchors.

Illustrates persistent Role entities (101) and their connections to:

- attributes (101a),
- relationships (101b),
- domain participation history (101c),
- and interpretability metadata (101d).

```
Role: Emma
- identity: R1
- attributes: {human, adult, female}
- relations: {sister = R3}
- participation: Domain: Conversation, Span: #14

Role: Lucy
- identity: R2
- attributes: {human, adult, female}
- relations: {sister = R4}
- participation: Domain: Conversation, Span: #14
```

FIG. 3

Domains as Meaning-Fields.

Shows Domains (102) as semantic environments constraining:

- primitive eligibility (120),
- role interactions (101 ↔ 102),
- interpretability thresholds (140),
- and Span formation (103).

```

Domain: Perception
  active roles: {Emma, Lucy}
  valid primitives: THEN, WHILE
  constraints: observer ≠ observed
  interpretability factors: shared perspective, distance

Domain: Cooperation
  active roles: {Emma, Lucy}
  valid primitives: WE, WHILE, UNTIL
  constraints: mutual intention
  interpretability factors: effort alignment, task proximity

```

FIG. 4

Span Lifecycle.

Depicts Span initiation (103a), maintenance (103b), and collapse (103c), triggered by:

- topic change,
- domain change (102 → 102'),
- speaker change,
- interpretability discontinuity (140),
- primitive lifecycle completion (120),
- or discourse markers.

```

[Span #12 – Conversation Domain]
  roles: Emma, Lucy
  content: "Emma told Lucy that her sister was arriving."
  primitives: THEN (link back), AmbiguityEngine active

Trigger: new information, topic shift

[Span #13 – Planning Domain]
  roles: Emma, Lucy
  content: "They decided to meet at the airport."
  primitives: WE (cooperative plan), UNTIL (goal condition)

```

DETAILED DESCRIPTION OF THE INVENTION

The following description provides a complete specification of the architecture, consistent with the figures and claims. This description is not limiting; variations within the defined scope are permitted.

7.1 CONCEPTUAL STATE REPRESENTATION

The architecture maintains a structured conceptual state comprising **Roles**, **Domains**, and **Spans**, each persistent within the continuous-present substrate (PEF) or a functionally equivalent environment.

These structures form the geometry of meaning — defining what exists (Roles), where meaning unfolds (Domains), and how reasoning is segmented (Spans).

7.1.1 Roles (101)

A **Role** is a persistent conceptual entity representing a participant in reasoning.

Roles correspond to:

- persons,
- objects,
- abstract constructs,
- collectives,
- hypothetical entities.

Role Properties

Each Role (101) includes:

- **Identity (101a):** persistent across spans;
- **Attributes (101b):** descriptive features;
- **Domain Participation History (101c):** record of semantic contexts;
- **Interpretability Metadata (101d):** alignment scores toward other entities;
- **Span Participation (101e):** involvement across conceptual episodes.

Role Persistence

Roles remain stable unless:

- primitive collapse detaches them,
- interpretability decay forces ejection,
- domain-switching requires reclassification.

Roles participate across Domains and Spans without losing identity.

Role Embodiments

Example 1 — Concrete Role

“Emma walked to the store.”

Role $R_1 = \textit{Emma}$ persists across domains.

Example 2 — Composite Role

“Emma and Lucy carried the table.”

WE primitive forms a composite Role (R_{12}).

Example 3 — Abstract Role

“The promise was broken.”

Role $R_5 = \textit{promise}$ participates in SocialContract Domain.

7.1.2 Domains (102)

A **Domain** is a semantic environment activated by verbs, modality, discourse cues, or contextual continuity.

Domains function as **meaning-fields** that dictate how Roles interact.

Domain Functions

Domains (102):

1. Determine **primitive eligibility** (120).
2. Constrain **Role interactions** ($101 \leftrightarrow 102$).
3. Govern **interpretability thresholds** (140).
4. Structure **Span formation and collapse** (103).

Domains effectively shape the semantic physics of reasoning.

Domain Activation

Domains may activate via:

- **Verb mapping** (carried → CooperativeTask, warned → Threat),
- **Modality cues** (might → Hypothetical),
- **Discourse cues** (“together”, “meanwhile”),
- **Continuity signals** (topic or agent consistency).

Domain Activation Logic (150) controls this process.

Domain Embodiments

Example 1 — Perception

“Emma looked at the horizon.” → Perception Domain.

Example 2 — Multi-Domain Layering

“Emma asked Lucy to help her carry the box.”

→ Conversation + Intention + CooperativeTask.

Example 3 — Domain Switching

“Emma laughed. The fire alarm sounded.”

→ Social → Threat → Span boundary.

7.1.3 Spans (103)

A **Span** is a bounded conceptual episode.

Spans regulate reasoning by:

- preventing drift,
- containing primitive effects,
- localizing transformations,
- enabling ambiguity persistence.

Spans are **non-temporal**; they represent conceptual segmentation, not time.

Span Lifecycle

Spans have three phases:

- **Initiation (103a):** triggered by new conceptual activity.
- **Maintenance (103b):** persists while stability remains.
- **Collapse (103c):** triggered by structural discontinuity.

See FIG. 4.

Span Boundary Conditions

The Span Boundary Algorithm (160) creates a new Span when:

- topic changes,
- domain shifts (102 → 102'),
- speaker changes,
- interpretability discontinuity occurs (140 < threshold),
- primitive lifecycle ends (120 collapses),
- discourse markers indicate transition ("then", "however").

Span Examples

Example 1 — Topic Change

"Emma cooked dinner. The fire alarm sounded."

→ New Span triggered.

Example 2 — Primitive Completion

"She waited until her sister arrived."

UNTIL collapses → new Span.

Example 3 — Domain Drift

"Emma hugged Lucy. Lucy stepped away."

Social → Movement → Span boundary.

7.1.4 Interaction Geometry of Roles, Domains, and Spans

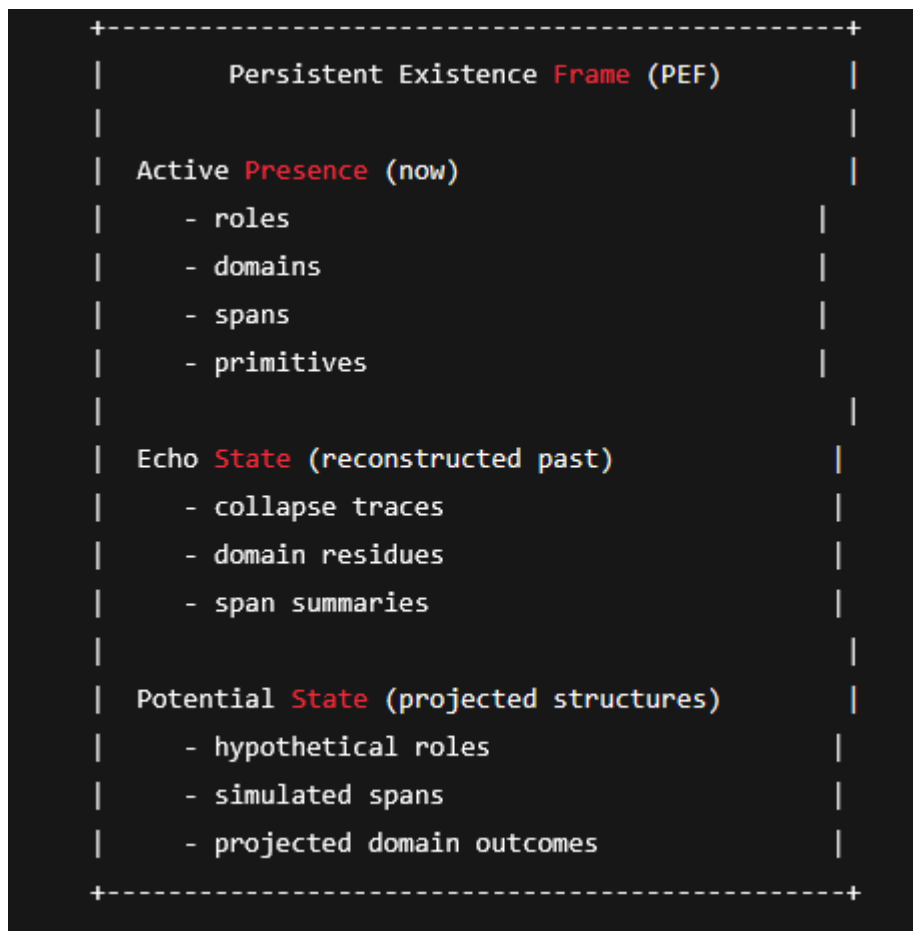


FIG. 6 — Interaction Geometry of Roles (101), Domains (102), and Spans (103).

These elements form a lattice of reasoning:

PEF (190)

└─ Span (103)

└─ Domain (102)

└─ Roles (101)

└─ Primitives (120)

- Roles inhabit Domains.
- Domains operate inside Spans.
- Spans live inside the continuous present.
- Primitives transform all three.

This geometry is the substrate for all operator-level reasoning.

7.2 COMPOSITIONAL PRIMITIVES (120)

The architecture's reasoning dynamics are governed by a family of **explicit compositional primitives**, each defined by:

- a semantic function,
- structural invariants,
- preconditions for activation,
- activation mechanics,
- transformation effects,
- collapse conditions,
- and embodiment examples.

These primitives transform Roles (101), Domains (102), and Spans (103) according to deterministic rules.

FIG. 5 illustrates how primitives interact with conceptual structures.

```
[Span: S1]
  roles: Emma
  domain: Perception
  primitive: THEN

      THEN (progression)

[Span: S2]
  roles: Emma, Lucy
  domain: Conversation
  primitive: WE, BECAUSE
```

7.2.1 WE — Primitive of Collective Formation (120a)

Semantic Function

WE forms a **composite Role (101f)** representing two or more Roles acting as a unified agent.

It encodes shared intention, joint action, or collective perspective.

Structural Invariants

- Constituent Roles maintain identity inside the composite.
- Interpretability between them must remain above threshold (140).
- Domain coherence must hold (typically CooperativeTask, JointAttention, Planning).
- Only one active WE per role-set per Span.

Preconditions for Activation

WE activates when:

- ≥ 2 Roles participate in a shared Domain (102),
- cooperative cues exist (e.g., “together”, “carried”),
- interpretability alignment is high (140),
- no prior incompatible WE structure exists in the Span.

Activation Mechanics

1. **Composite Role Creation**
Instantiate composite Role (101f).
2. **Domain Calibration**
Strengthen CooperativeTask or Planning Domains.
3. **Interpretability Harmonization**
Combine participant interpretability metadata.
4. **Span Attachment**
Attach WE to the active Span (103).

Transformation Effects

- Composite Role (101f) becomes available for subsequent primitives.

- Obligations or causal effects attach to the composite rather than individual Roles.
- Future domain activations may treat the composite as a single agent.

Collapse Conditions

WE collapses when:

- interpretability between constituent Roles drops ($140 < \text{threshold}$),
- a domain switch invalidates collective action,
- discourse signals separation (“Emma left Lucy”),
- or Span ends.

Embodiment Examples

Example 1 — Cooperative Action

“Emma and Lucy carried the table.”

→ WE activates, composite Role (101f).

Example 2 — Joint Perspective

“We decided to stay inside.”

→ WE encodes shared deliberation.

7.2.2 THEN — Primitive of Conceptual Progression (120b)

Semantic Function

THEN creates a directed conceptual progression between Spans.

It encodes ordered succession without temporal dependency.

Structural Invariants

- Preceding Span (103) and succeeding Span (103') must both exist.
- Conceptual continuity must be coherent.
- THEN-links cannot form cycles.

Preconditions for Activation

THEN activates when:

- discourse indicates progression (“then”, “afterward”),
- domain changes necessitate a new Span,
- or primitive collapse forces continuation.

Activation Mechanics

1. **Span Creation**
Create new Span (103’).
2. **Successor Link**
Establish directed edge: $\text{Span}_1 \rightarrow \text{Span}_2$.
3. **Context Inheritance**
Copy Roles and Domains unless replaced.

Transformation Effects

- Enables structured conceptual chains.
- Allows causal primitives to operate across spans.
- Creates the non-temporal “flow” of reasoning.

Collapse Conditions

THEN collapses when:

- spans merge due to reinterpretation,
- domain drift invalidates the linkage,
- contradictory primitives activate.

Embodiment Example

“Emma opened the door. Then she stepped outside.”

→ New Span created; Movement Domain activated.

7.2.3 WHILE — Primitive of Conditional Continuity (120c)

Semantic Function

WHILE maintains a conceptual environment *as long as a condition holds*.

It encodes persistent states, background activity, or overlapping processes.

Structural Invariants

- Condition must evaluate true at activation.
- Condition is tested continuously.
- Collapse must occur immediately on failure.

Preconditions for Activation

WHILE activates when:

- a conditional is explicit or implied,
- Roles maintain stable interpretability (140),
- Domain supports continuity (e.g., Cooking, Movement, Emotion).

Activation Mechanics

1. Sub-Span Creation

WHILE opens a persistent sub-Span (103w).

2. Condition Monitoring

Aurora Block evaluates collapse triggers at each cycle.

Transformation Effects

- Encodes background context.
- Supports nested reasoning.
- Maintains environment stability.

Collapse Conditions

WHILE collapses when:

- condition becomes false,
- domain switch contradicts continuity,
- interpretability drop undermines structure,
- a superseding primitive activates.

Embodiment Example

“Emma hummed while she cooked dinner.”

→ WHILE maintains sub-Span for the cooking process.

7.2.4 UNTIL — Primitive of Goal-Bound Persistence (120d)

Semantic Function

UNTIL persists a conceptual state until a target condition is met.
It encodes goal-directed processes.

Structural Invariants

- Target condition must be structurally representable.
- Progress must be trackable across updates.

Preconditions for Activation

UNTIL activates when:

- a goal or endpoint is indicated,
- domain supports directed progression (Planning, Movement),
- role stability allows sustained evaluation.

Activation Mechanics

1. Goal Span Initiation

Create persistent Span (103u).

2. Progress Evaluation

Monitor alignment between conceptual state and target condition.

Transformation Effects

- Encodes waiting, striving, or procedural continuation.
- Manages long-range conceptual coherence.

Collapse Conditions

UNTIL collapses when:

- target condition becomes true,
- goal becomes impossible,
- domain drift prevents continuation.

Embodiment Example

“She waited until her sister arrived.”

→ UNTIL collapses upon arrival, transitioning to new Span.

7.2.5 BECAUSE — Primitive of Causal Attribution (120e)

Semantic Function

BECAUSE creates a **causal link** between a cause Span (103c) and an effect Span (103e).

It encodes structural, not temporal, causation.

Structural Invariants

- Cause and effect must be valid spans.
- Interpretability between causal elements must exceed threshold.
- Loops are prohibited.

Preconditions for Activation

BECAUSE activates when:

- causal signals present (“because”, “so that”),
- explanatory need detected,
- structural alignment supports dependency.

Activation Mechanics

1. **Span Pair Selection**
Identify cause Span and effect Span.
2. **Causal Link Formation**
Attributed as BECAUSE: $\text{Span}_1 \rightarrow \text{Span}_2$.
3. **Alignment Adjustment**
Update interpretability metadata (140).

Transformation Effects

- Supports explanation, justification, and reconstruction.
- Provides anchor points for PEF Echo State.

Collapse Conditions

BECAUSE collapses when:

- alternative cause emerges,
- contradiction appears,
- causal dependency becomes incoherent.

Embodiment Example

“Emma left early because she felt unwell.”

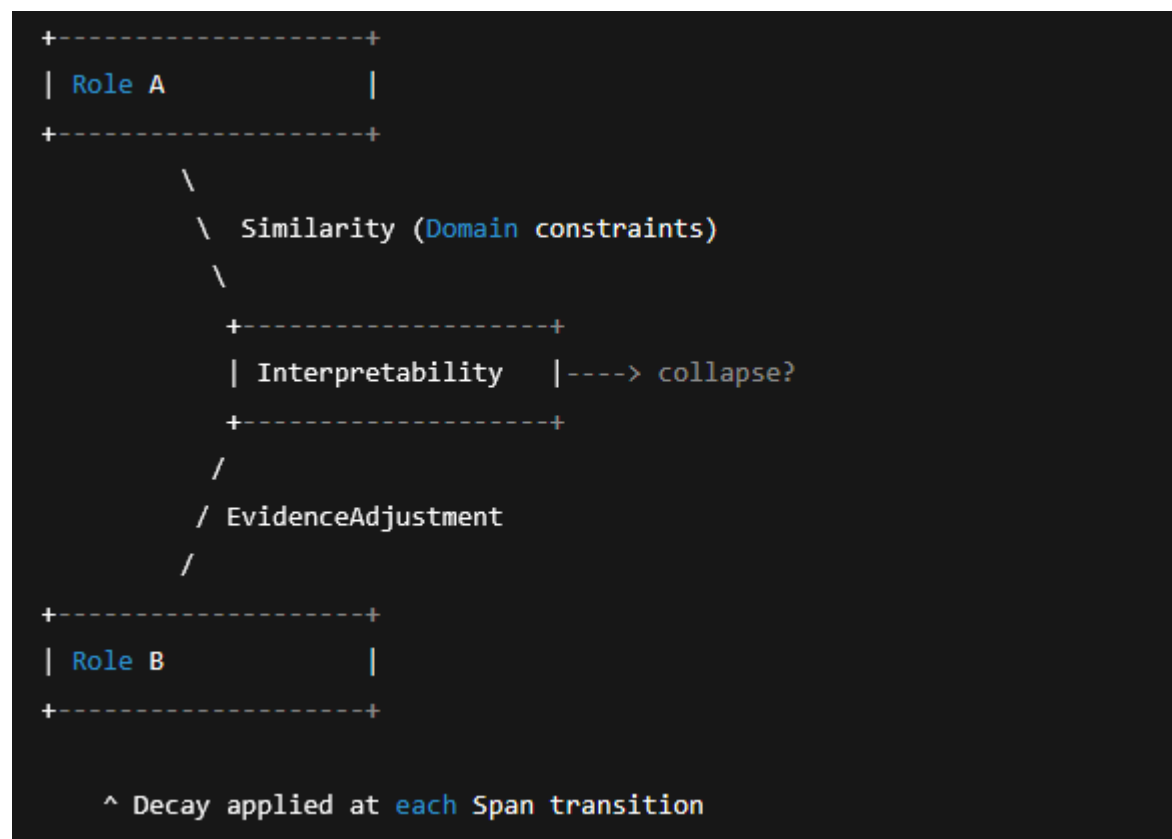
→ BECAUSE maps cause to effect via linked Spans.

7.3 INTERPRETABILITY ENGINE (140)

The Interpretability Engine evaluates conceptual coherence between entities in the active reasoning state. It governs:

- Role compatibility
- Domain stability
- Primitive persistence or collapse
- Span continuation or boundary creation
- Ambiguity maintenance and resolution

FIG. 7 illustrates the interpretability computation.



7.3.1 Interpretability Formula (140a)

Interpretability I between conceptual entities is computed as:

$$I = S + E - D$$

Where:

- **S — Similarity:**
Structural similarity between entities (shared attributes, domain participation, interaction patterns).
- **E — Evidence:**
Contextual reinforcement derived from discourse, primitive activation, continuity, or repeated co-occurrence.
- **D — Decay:**
A dynamic term representing conceptual drift.
Decay increases when:
 - roles diverge in domain participation,
 - competing structures emerge,
 - an entity is unused across multiple spans.

A primitive, Domain, or Span collapses when $I < \text{threshold}$.

7.3.2 Interpretability Domains

Interpretability operates in three layers:

1. **Role–Role Interpretability (140b)**
Determines whether WE, THEN, or BECAUSE may operate.
2. **Role–Domain Interpretability (140c)**
Determines whether a Role remains coherent within a meaning-field (102).
3. **Span–Span Interpretability (140d)**
Used for reconstruction in the PEF and for evaluating conceptual succession.

7.3.3 Collapse Triggers (140e)

Interpretability collapse may occur due to:

- Low alignment between Roles,
- Domain incompatibility,
- Primitive contradiction,
- abrupt domain switching,
- parallel interpretation divergence.

Collapse may trigger:

- primitive termination,
- span boundary formation,
- domain reclassification,
- or removal of an interpretation from the parallel set.

7.3.4 Embodiments

Example 1 — WE Collapse

“Emma and Lucy carried the table. Emma left early.”

Role alignment drops → WE collapses.

Example 2 — Domain Drift

“Emma warned Lucy. Lucy laughed.”

Threat → Social Domain → Span boundary.

Example 3 — Ambiguity Persistence

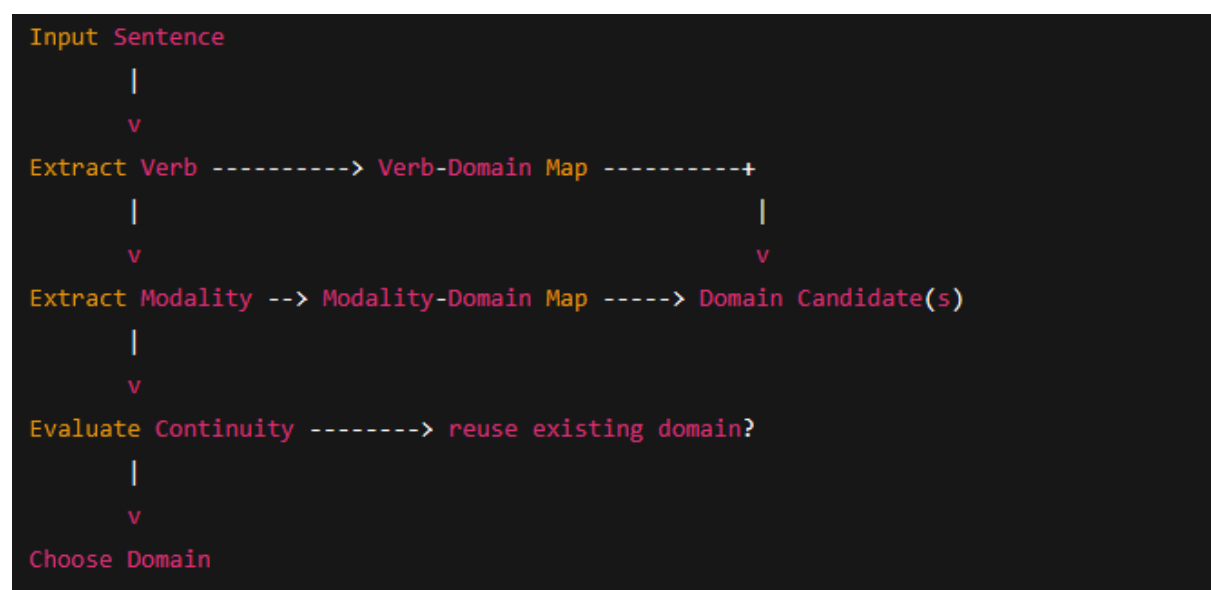
“Emma told Lucy that her sister was arriving.”

Both interpretations persist until context modifies interpretability.

7.4 DOMAIN ACTIVATION LOGIC (150)

Domain Activation Logic identifies and manages semantic meaning-fields that regulate allowable primitive interactions and conceptual structure.

FIG. 8 illustrates domain activation pathways.



7.4.1 Activation Inputs

Domains activate based on:

1. Verb Mapping (150a)

- carried → CooperativeTask
- warned → Threat
- decided → Planning
- noticed → Perception

2. Modality Cues (150b)

- might, could → Hypothetical
- must, should → Obligation
- intended to → Intention

3. Continuity Signals (150c)

- persistent Roles
- stable topics
- ongoing primitive structure

4. Discourse Cues (150d)

- “together”, “again”, “meanwhile”, “because”

7.4.2 Domain Lifecycle

Domains proceed through:

- **Activation (102a)**
- **Stabilization (102b)**
- **Layering (102c)**
- **Drift or Conflict (102d)**
- **Collapse (102e)**

A domain collapses when:

- interpretability drops,
- contradictory primitive activates,

- domain incompatibility arises,
- or a Span boundary forms.

7.4.3 Primitive–Domain Compatibility

Each primitive requires specific domain conditions:

- **WE (120a)** → CooperativeTask / Planning
- **THEN (120b)** → broad compatibility
- **WHILE (120c)** → continuous, stable domain
- **UNTIL (120d)** → Planning / Intention
- **BECAUSE (120e)** → explanatory or evidentiary domain

Domain constraints prevent illegal state transitions.

7.4.4 Embodiments

Example 1 — Planning Context

“She intended to call Lucy.”

→ Planning Domain activates → UNTIL eligible.

Example 2 — Domain Switching

“Emma laughed. The fire alarm sounded.”

→ Social → Threat → new Span.

Example 3 — Multi-Domain Layering

“Emma asked Lucy to help carry the box.”

→ Conversation + Intention + CooperativeTask.

7.5 SPAN BOUNDARY ALGORITHM (160)

The Span Boundary Algorithm determines when a conceptual episode ends and a new one begins.

FIG. 9 illustrates boundary logic.

```

[Input Stream]
|
v
Topic Similarity -----> NO ---> New Span
|
v YES
Domain Change -----> YES --> New Span
|
v NO
Speaker Change -----> YES --> New Span
|
v NO
Interpretability Shift -> YES --> New Span
|
v NO
Goal Completion -----> YES --> New Span
|
v NO
Maintain Span

```

7.5.1 Purpose

Span boundaries prevent:

- conceptual drift,
- primitive interference across episodes,
- domain contamination,
- premature collapse,
- interpretability degradation.

They support ambiguity maintenance and structured reasoning.

7.5.2 Boundary Conditions (160a)

A new Span (103') is created when any of the following occur:

1. **Topic Change**
Shift in subject matter or event type.

2. **Domain Change (102 → 102')**
E.g., Social → Threat.
3. **Speaker Change**
Essential in dialogue.
4. **Interpretability Discontinuity (140)**
Alignment drops below threshold.
5. **Primitive Lifecycle Completion (120)**
UNTIL completes; WHILE condition fails.
6. **Discourse Markers**
“Then”, “However”, “Suddenly”.
7. **Ambiguity Collapse**
Parallel interpretations reduce asymmetrically.

7.5.3 Span Continuation (160b)

A Span persists when:

- domain remains stable,
- interpretability remains above threshold,
- primitives remain active or stable,
- discourse maintains cohesion.

7.5.4 Span Collapse Mechanics (160c)

Collapse results in:

- Echo traces stored in PEF,
- domain realignment,
- updated primitive availability,
- possible causal inference (BECAUSE),
- potential activation of THEN for succession.

7.5.5 Embodiments

Example 1 — Topic Change

“Emma cooked dinner. The fire alarm sounded.”

→ Span boundary.

Example 2 — Primitive Completion

“She waited until her sister arrived.”

→ UNTIL collapses → new Span created.

Example 3 — Domain Drift

“Emma hugged Lucy. Lucy stepped away.”

→ Social → Movement → boundary.

7.6 PARALLEL INTERPRETATION ENGINE (170)

The Parallel Interpretation Engine maintains **multiple structurally valid interpretations** simultaneously, resolving ambiguity only when structural constraints force collapse.

This capability distinguishes the system from statistical or heuristic models which collapse ambiguity prematurely.

FIG. 10 illustrates the interpretation set mechanics.

```
Define Semantic Function
|
v
Define Invariants -----> Invariants guide activation & collapse
|
v
Boundary Conditions -----> Inside/Outside concept space
|
v
Activation Rules -----> Preconditions for instantiating primitive
|
v
Transformation Effects ----> Structural update inside PEF
|
v
Collapse Conditions -----> How primitive ends and returns structure to PEF
```

7.6.1 Purpose

The Parallel Interpretation Engine enables the system to:

- maintain ambiguity where appropriate,
- evaluate competing meaning configurations,
- prevent premature collapse,
- reconcile interpretations when evidence strengthens one path,
- eliminate interpretations when structure forbids them.

Interpretations exist in parallel **within the continuous present**, not as temporal alternatives.

7.6.2 Interpretation Sets (170a)

At any moment, the system maintains a set of interpretations:

$$I = \{ i_1, i_2, \dots, i_n \}$$

Each interpretation includes:

- Role bindings (101 $\leftrightarrow$ 101')
- Domain configurations (102)
- Span structures (103)
- Active primitives (120)
- Interpretability metadata (140)
- PEF state contributions (190)

7.6.3 Activation

A new interpretation set is generated when:

- ambiguity is detected,
- role binding is non-unique (e.g., pronoun),
- domain cues promote multiple readings,
- causal relations can be assigned in more than one way.

7.6.4 Maintenance

Interpretations persist while:

- structural validity holds,
- interpretability remains above threshold,
- no primitive contradicts the configuration,
- domain constraints remain satisfied.

Each interpretation undergoes the full Aurora Block cycle (180) independently.

7.6.5 Collapse Rules (170b)

Collapse occurs when:

1. **Primitive Incompatibility**
A primitive activates in one interpretation that cannot activate in another.
2. **Interpretability Asymmetry**
If I_i drops below threshold while I_j does not.
3. **Domain Constraint Violation**
Structure incompatible with domain invariants.
4. **Contextual Evidence**
New information favors one interpretation exclusively.
5. **Span Incompatibility**
Spans diverge beyond reconcilable form.

After collapse, only structurally valid interpretations remain.

7.6.6 Embodiment Example — Pronoun Ambiguity

Input:

“Emma told Lucy that her sister was arriving.”

Two interpretations:

- i_1 : “her” = Emma’s sister
- i_2 : “her” = Lucy’s sister

Both persist.

Follow-up:

“Emma’s sister had emailed earlier.”

→ i_2 collapses, i_1 remains.

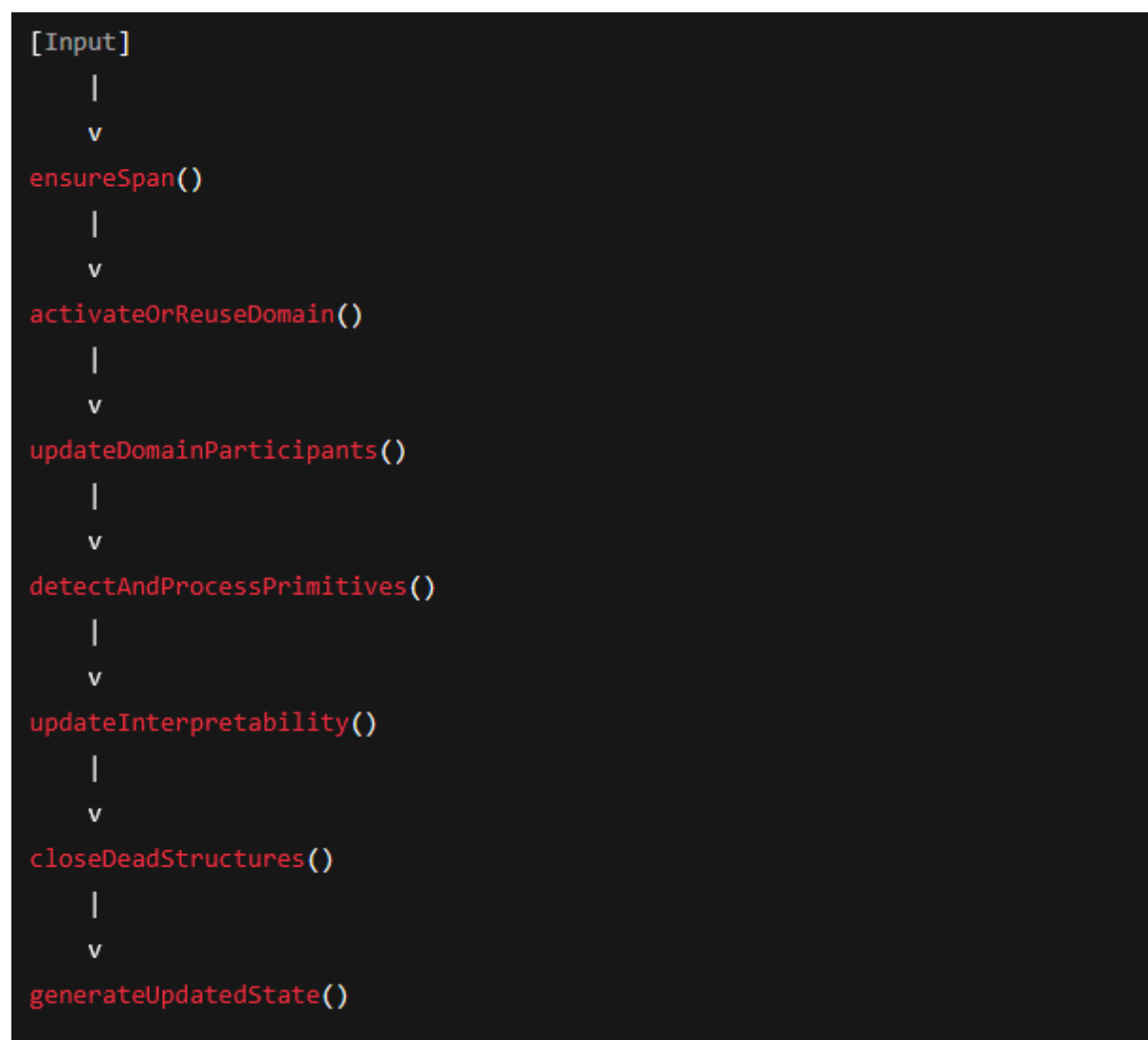
This demonstrates non-temporal, structure-first ambiguity resolution.

7.7 AURORA PRIMITIVE DERIVATION METHOD (APDM) (175)

APDM provides the formal generative procedure for defining compositional primitives.

It ensures all primitives are structured, interoperable, and fully examinable.

FIG. 11 illustrates APDM stages.



7.7.1 Purpose of APDM

APDM guarantees that each primitive has:

- defined semantics,
- explicit invariants,
- clear boundaries,
- deterministic activation,
- controlled collapse,
- structural compatibility with all other primitives.

APDM is essential to the invention because it enables future expansion without redesign.

7.7.2 APDM Six-Stage Derivation Process

Each primitive (120) is defined using:

1. Semantic Function (175a)

What conceptual transformation the primitive performs.

2. Structural Invariants (175b)

What must remain true while the primitive is active.

3. Boundary Definition (175c)

Entry, persistence, and exit conditions.

4. Activation Rules (175d)

What triggers the primitive.

5. Transformation Effects (175e)

What changes the primitive induces in Roles (101), Domains (102), Spans (103).

6. Collapse Conditions (175f)

What terminates the primitive.

This uniform schema ensures internal consistency and system extensibility.

7.7.3 Example: Application to WE Primitive

Semantic: unify roles

Invariants: shared domain and high interpretability

Boundary: starts with cooperative cue

Activation: roles align
 Effects: composite Role formed
 Collapse: interpretability drops or roles diverge

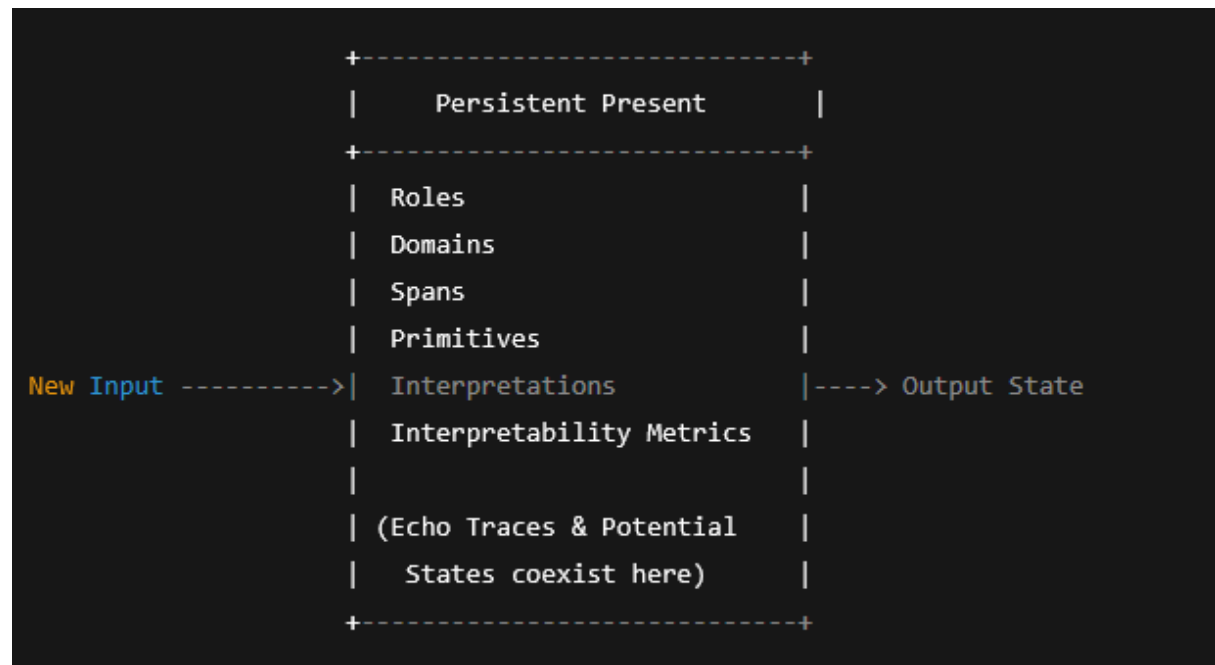
This illustrates APDM's clarity and necessity.

7.8 AURORA BLOCK EXECUTION MODULE (180)

The Aurora Block is the system’s **deterministic reasoning engine**.
 It executes the complete reasoning cycle:

- interpreting inputs,
- updating conceptual state,
- managing primitives,
- controlling collapse,
- maintaining interpretations,
- synthesizing final output.

FIG. 12 illustrates the execution flow.



7.8.1 Purpose

The Aurora Block:

- orchestrates all reasoning processes,
- enforces structural constraints,
- integrates Domains, Spans, and Roles,
- preserves ambiguity,
- initiates collapse when required,
- ensures deterministic compositional updates.

It is the architecture's computational center.

7.8.2 Deterministic Processing Cycle (180a)

The cycle contains nine steps:

Step 1 — Interpret Input

Convert input into Roles (101), Domains (102), Spans (103), and candidate primitives (120).

Step 2 — Evaluate Span Boundaries

Apply boundary algorithm (160).

Step 3 — Activate or Update Domains

Apply Domain Activation Logic (150).

Step 4 — Update Interpretability

Compute $I = S + E - D$.

Step 5 — Evaluate Primitive Eligibility

Check structural invariants (175b) and domain compatibility.

Step 6 — Apply Primitives

Modify Roles, Domains, and Spans accordingly.

Step 7 — Evaluate Collapse Conditions

Detect primitive, domain, or interpretability collapse.

Step 8 — Update Parallel Interpretation Set

Remove incompatible interpretations; preserve valid ones.

Step 9 — Emit Updated Conceptual State

Output the new state for downstream processing.

7.8.3 Embodiments

A minimal executable embodiment is a pronoun-disambiguation engine that:

- creates Roles,
- activates Domains,
- updates interpretability,
- maintains parallel interpretations,
- collapses ambiguity as evidence appears.

This demonstrates end-to-end enablement.

7.9 PERSISTENT EXISTENCE FRAME (PEF) INTEGRATION (190)

The PEF is *not* redefined here.

It is fully described in earlier priority documents.

This section explains **how Aurora operates within the PEF**, using its continuous-present substrate.

7.9.1 PEF Structural States

Aurora interacts with three PEF states:

Active Presence (190a)

Where Roles, Domains, Spans, and primitives currently exist.

Echo State (190b)

Where collapsed spans and prior structures leave reconstructive traces.

Potential State (190c)

Where hypothetical or projected states exist (e.g., alternative interpretations).

7.9.2 Non-Temporal Operation

Reasoning occurs **without temporal modeling**:

- Spans represent conceptual—not chronological—episodes.

- Reconstruction uses Echo traces, not time.
- Projection uses Potential State, not predictions.
- Primitive cycles operate independent of clocks.

7.9.3 Aurora Block in PEF Environment

The Aurora Block uses PEF states for:

- boundary evaluation (Echo),
- interpretability calibration (Potential),
- ambiguity management (Active + Echo + Potential),
- reconstruction and projection simultaneously.

7.9.4 Embodiments

Reconstruction Example

“Emma left. She apologized later.”

Echo traces enable correct antecedent recovery.

Projection Example

“Emma might call Lucy.”

Potential State holds alternate futures until grounded.

CLAIMS

Claim 1 (Independent System Claim)

A machine reasoning system comprising:

1. **a structured conceptual state** including:
 - a plurality of Roles representing persistent conceptual entities (101),
 - one or more Domains representing semantic meaning-fields (102), and
 - one or more Spans representing bounded conceptual episodes (103);
2. **a plurality of compositional primitives (120)**, each primitive defined by:
 - a semantic function,
 - structural invariants,
 - preconditions for activation,
 - transformation effects on Roles, Domains, and Spans, and
 - collapse conditions;
3. **an interpretability engine (140)** configured to compute interpretability values between conceptual entities according to similarity, contextual evidence, and conceptual decay;
4. **domain activation logic (150)** configured to activate or modify Domains in response to verbs, modalities, discourse cues, continuity signals, or role participation patterns;
5. **a span boundary algorithm (160)** configured to determine whether to continue or terminate a Span based on topic change, domain change, speaker change, interpretability discontinuity, primitive lifecycle completion, or discourse markers;
6. **a parallel interpretation engine (170)** configured to instantiate and maintain multiple structurally valid interpretations of input concurrently and collapse incompatible interpretations when required by structural constraints; and
7. **a deterministic processing module (180)** configured to:
 - interpret input,
 - evaluate span boundaries,

- activate or update Domains,
- compute interpretability values,
- evaluate primitive eligibility,
- activate primitives,
- evaluate collapse conditions,
- update the set of parallel interpretations, and
- output an updated conceptual state.

Claim 2 (Roles)

The system of claim 1 wherein each Role includes identity properties, attributes, interpretability metadata, domain participation history, and span participation history.

Claim 3 (Domains)

The system of claim 1 wherein each Domain constrains primitive eligibility, interpretability thresholds, and allowable role interactions.

Claim 4 (Spans)

The system of claim 1 wherein each Span represents a conceptual episode whose lifecycle is governed by domain stability, interpretability thresholds, primitive activity, and discourse markers.

Claims 5–9: Primitives

Claim 5 — WE Primitive

The system of claim 1 wherein the compositional primitives include a WE primitive configured to instantiate a composite Role from multiple Roles when cooperative or coordinated activity is present, and to collapse the composite Role when interpretability between constituent Roles falls below a threshold or a domain shift occurs.

Claim 6 — THEN Primitive

The system of claim 1 wherein the compositional primitives include a THEN primitive configured to generate a new Span and a directed conceptual progression from a preceding Span.

Claim 7 — WHILE Primitive

The system of claim 1 wherein the compositional primitives include a WHILE primitive configured to maintain a persistent Span while a condition remains satisfied and to collapse the Span when the condition becomes false or a domain changes.

Claim 8 — UNTIL Primitive

The system of claim 1 wherein the compositional primitives include an UNTIL primitive configured to maintain a Span until a target condition is satisfied and to collapse the Span upon detection of the target condition or domain incompatibility.

Claim 9 — BECAUSE Primitive

The system of claim 1 wherein the compositional primitives include a BECAUSE primitive configured to form a causal relationship between a cause Span and an effect Span and to collapse the causal link upon contradiction or resolution.

Interpretability & Ambiguity Claims: 10–13

Claim 10 — Interpretability Formula

The system of claim 1 wherein interpretability values are computed according to:
 $I = S + E - D$,
where S represents similarity, E represents contextual evidence, and D represents conceptual decay.

Claim 11 — Interpretability Collapse

The system of claim 10 wherein collapse of a Role, Domain, or Span occurs when the interpretability value falls below a threshold.

Claim 12 — Ambiguity Maintenance

The system of claim 1 wherein multiple interpretations are maintained concurrently until domain constraints, interpretability decay, primitive incompatibility, or contextual evidence remove one or more interpretations.

Claim 13 — Pronoun Ambiguity Embodiment

The system of claim 1 wherein ambiguity in pronoun antecedent resolution is preserved across multiple interpretations until structural or contextual constraints collapse competing interpretations.

Domain & Span Claims: 14–16

Claim 14 — Verb-Based Domain Activation

The system of claim 1 wherein verbs in an input activate Domains corresponding to semantic meaning-fields.

Claim 15 — Multi-Domain Layering

The system of claim 1 wherein multiple Domains may be active simultaneously when discourse or conceptual structure indicates overlapping semantic environments.

Claim 16 — Span Boundary Creation

The system of claim 1 wherein a new Span is created upon detection of domain change, topic change, speaker change, primitive collapse, interpretability discontinuity, or discourse markers.

APDM Claims: 17–18

Claim 17 — APDM Generative Method

The system of claim 1 wherein each compositional primitive is defined according to a Primitive Derivation Method comprising:

1. semantic function definition,
2. invariant specification,
3. boundary definition,
4. activation rule formation,
5. transformation effect specification, and
6. collapse condition specification.

Claim 18 — Primitive Extensibility

The system of claim 17 wherein new primitives can be added without modification to existing primitives or the processing module.

Aurora Block Claims: 19–20

Claim 19 — Deterministic Reasoning Cycle

The system of claim 1 wherein the processing module executes a deterministic reasoning cycle comprising interpretation, span evaluation, domain activation, interpretability update, primitive evaluation, primitive activation, collapse evaluation, and output generation.

Claim 20 — Non-Statistical Reasoning

The system of claim 19 wherein primitive activation and collapse are governed by rule-based constraints rather than statistical inference.

PEF Integration Claims: 21–23

Claim 21 — Operation Within PEF

The system of claim 1 operating within a continuous-present computational substrate comprising Active Presence, Echo State, and Potential State.

Claim 22 — Reconstruction and Projection

The system of claim 21 wherein prior conceptual structures are reconstructed using Echo State and hypothetical structures are represented using Potential State.

Claim 23 — Non-Temporal Sequence

The system of claim 21 wherein conceptual succession is represented by Span ordering independent of temporal encoding.

Independent Method Claim: 24

Claim 24 (Independent Method Claim)

A method for performing compositional reasoning, comprising:

- maintaining Roles, Domains, and Spans;
- computing interpretability values;
- activating compositional primitives;
- transforming conceptual structures;
- collapsing primitives, Domains, or Spans when collapse conditions are met;

- maintaining multiple interpretations concurrently;
- and outputting an updated conceptual state.

Machine/Software Claims: 25–26

Claim 25 — Software Implementation

The system of claim 1 implemented in software, hardware, or hybrid computational architecture.

Claim 26 — Data Structure Claim

A non-transitory computer-readable medium storing data structures representing Roles, Domains, Spans, primitives, interpretability values, and conceptual succession relationships enforceable by the processing module.

Final Catch-All Claim: 27

Claim 27 — Embodiments

A system or method substantially as described herein with reference to any embodiment, figure, or combination of embodiments.